COPV Blowdown to Propellant Tanks: Governing Equations for Sizing

Pressurization system analysis

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Abstract

This note sets out the governing equations and solution approach for sizing the GN_2 composite overwrapped pressure vessel (COPV) that pressurizes the LOx and IPA tanks through a single regulator. The aim is the minimum COPV volume and the required pressure rating. The model is real-gas throughout, includes Joule–Thomson (JT) cooling across the regulator, neglects line pressure losses, and uses constant propellant mass flow rates with a constant instantaneous collapse factor in each tank. The COPV also pre-pressurizes both ullages. No numerical values are evaluated here.

1 System and phases

The pressurant path is COPV → regulator (isenthalpic) → both propellant tank ullages. Line pressure losses are neglected, so the regulator outlet pressure equals the tank pressure. Both tanks sit behind one regulator, so they see the same pressure P_t and the same inlet temperature T_{in}.

The model runs in two phases:

- **Phase 1, pre-pressurization:** fill both ullages, and the downstream line volume, from their initial pressure up to P_t.
- **Phase 2, expulsion:** hold P_t while propellant flows out at constant rates.

The unknowns are the COPV volume V_b and its initial pressure P_{b0}. The initial temperature T_{b0} is known. The pressure rating follows from P_{b0} (Section 6).

2 COPV (bottle) equations

The bottle is a rigid control volume with one outlet. Conservation of mass and energy:

$$\frac{\mathrm{d}m_b}{\mathrm{d}t} = -\dot{m}, \qquad \frac{\mathrm{d}(m_b u_b)}{\mathrm{d}t} = -\dot{m}\, h_b + \dot{Q}_w, \tag{1}$$

where $\dot{m}$ is the total pressurant flow to both tanks. Substituting the mass equation and $h_b - u_b = P_b v_b$ gives

$$m_b \frac{\mathrm{d}u_b}{\mathrm{d}t} = -\dot{m}\, P_b v_b + \dot{Q}_w. \tag{2}$$

Writing $u = u(T, v)$ with the real-gas identity

$$\mathrm{d}u = c_v\, \mathrm{d}T + \left[T \left(\frac{\partial P}{\partial T} \right)_v - P \right] \mathrm{d}v, \tag{3}$$

and using $v_b = V_b/m_b$ (so $\mathrm{d}v_b/\mathrm{d}t = v_b\,\dot{m}/m_b$), gives the bottle temperature equation:

$$\boxed{m_b c_v \frac{\mathrm{d}T_b}{\mathrm{d}t} = -\dot{m}\, v_b\, T_b \left(\frac{\partial P}{\partial T}\right)_v + \dot{Q}_w} \tag{4}$$

For an ideal gas, $T(\partial P/\partial T)_v = P$, and Eq. (4) collapses to the familiar $mc_v\,\mathrm{d}T/\mathrm{d}t = -\dot{m}RT$.

2.1 Adiabatic case

With $\dot{Q}_w = 0$ (the sizing default: a short blowdown, and the composite overwrap is a poor conductor), the equations reduce exactly to constant entropy:

$$s_b(t) = s(P_{b0}, T_{b0}) \equiv s_0, \qquad \rho_b(t) = \frac{m_{b0} - m_{used}(t)}{V_b}. \tag{5}$$

The bottle state at any instant then depends only on how much mass has been removed, not on the flow rate or the history: $P_b = P(\rho_b, s_0)$ and $T_b = T(\rho_b, s_0)$. This makes the pre-press rate and the hold time irrelevant for the bottle.

> Assuming no heat return during the hold between pre-press and run is conservative. In reality the bottle gas re-warms toward ambient, which raises P_b.

2.2 Optional heat-transfer bound

Use a lumped liner/overwrap wall node:

$$\dot{Q}_w = h_i A_i (T_w - T_b), \qquad m_w c_w \frac{\mathrm{d}T_w}{\mathrm{d}t} = -h_i A_i (T_w - T_b), \tag{6}$$

with h_i from a natural-convection correlation inside the bottle. An isothermal bottle is the opposite bound and gives the smallest volume.

2.3 Equation of state

The system is closed with a real-fluid equation of state for nitrogen [2], evaluated through CoolProp [3] or REFPROP, not $PV = mRT$. At bottle pressures the compressibility factor Z differs noticeably from 1.

3 Regulator and the Joule–Thomson effect

Treat the regulator as a steady-flow throttle with no work, no heat transfer and negligible kinetic-energy change. Enthalpy is then conserved across it:

$$\boxed{h_{in}(t) = h_b(t) \quad \Rightarrow \quad T_{in}(t) = T\big(P_t,\ h_b(t)\big)} \tag{7}$$

The JT coefficient describes the same process in differential form:

$$\mu_{JT} = \left(\frac{\partial T}{\partial P}\right)_h = \frac{1}{c_p}\left[T\left(\frac{\partial v}{\partial T}\right)_P - v\right], \qquad T_{in} = T_b - \int_{P_t}^{P_b} \mu_{JT}\,\mathrm{d}P\Big|_h. \tag{8}$$

A linearized form, $T_{in} \approx T_b - \bar{\mu}_{JT}(P_b - P_t)$, is useful as a hand check. In the model, the exact isenthalpic relation (7) should be used, because μ_{JT} varies strongly with both T and P over the blowdown.

Two cooling effects stack: the bottle gas cools as it expands isentropically, so T_b falls; the throttle then cools the gas further. T_{in} therefore falls throughout the run, $\rho_{in} = \rho(P_t, T_{in})$ rises, and each litre of displaced propellant costs more pressurant mass as the bottle empties. This is the main coupling between the bottle size and the demand.

During pre-press the regulator outlet pressure is the rising ullage pressure $P_u(t)$, not P_t, so in that phase $T_{in} = T(P_u(t), h_b)$: a larger pressure drop with more JT cooling.

4 Propellant tank demand (expulsion phase)

For each tank i (LOx, IPA) the outflow mass rate $\dot{m}_{p,i}$ is constant, so the volumetric displacement rate is

$$\dot{V}_i = \frac{\dot{m}_{p,i}}{\rho_{p,i}}, \qquad V_{u,i}(t) = V_{u0,i} + \dot{V}_i\, t. \tag{9}$$

The ideal pressurant flow fills the displaced volume with gas at inlet conditions, with no heat or mass transfer; this is the basis of the collapse-factor definition reviewed by de Quay and Hodge [1]. With a constant instantaneous collapse factor:

$$\boxed{\dot{m}_{g,i}(t) = CF_i\ \rho\big(P_t,\ T_{in}(t)\big)\ \dot{V}_i} \tag{10}$$

$$\dot{m}(t) = \sum_i \dot{m}_{g,i}(t)\, \mathbb{1}[t < t_{dep,i}]\,, \qquad t_{dep,i} = \frac{m_{p0,i} - m_{res,i}}{\dot{m}_{p,i}}. \tag{11}$$

The indicator function switches off each tank's demand when it depletes; the two tanks generally empty at different times. Because T_{in} is evaluated at each instant, the JT and bottle cooling enter the ideal mass directly, and the CF represents only the heat- and mass-transfer collapse inside the tank. This is consistent with the CF definition used earlier in this project.

If P_t is allowed to vary (for example with a regulator droop law), the term $V_{u,i}\,\mathrm{d}\rho_u/\mathrm{d}t$ for repressurizing the existing ullage must be added. With P_t constant it vanishes.

5 Pre-pressurization phase

Each ullage, plus its share of the downstream line volume V_{line}, is a rigid volume filled with gas at enthalpy $h_b(t)$:

$$\frac{\mathrm{d}m_{u,i}}{\mathrm{d}t} = \dot{m}_{pp,i}, \qquad \frac{\mathrm{d}(m_{u,i}u_{u,i})}{\mathrm{d}t} = \dot{m}_{pp,i}\, h_b(t) + \dot{Q}_{u,i}. \tag{12}$$

For the adiabatic ideal fill, integration gives the final ullage state:

$$u_{u,f} = \frac{m_{u0}u_{u0} + \int \dot{m}_{pp}\, h_b\, \mathrm{d}t}{m_{u0} + \Delta m_{pp}}, \qquad P(\rho_{u,f}, u_{u,f}) = P_t. \tag{13}$$

The final internal energy exceeds the inlet enthalpy because of flow work, so the adiabatic fill ends hot and needs the least mass. Real pre-press loses that heat to the cold wall and the propellant surface, so a pre-press collapse factor is applied:

$$m_{pp} = \sum_i CF_{pp,i}\ \Delta m_{pp,i}^{ideal} + \Delta m_{line}. \tag{14}$$

A conservative upper bound that avoids choosing a factor is to assume the gas ends at the tank wall or propellant temperature, which gives the densest ullage. In practice this phase is integrated in a few steps, because h_b falls as mass leaves the bottle; alternatively, iterate on the end state, which is path-independent for the adiabatic bottle.

6 Closing the sizing problem

6.1 Mass budget

$$m_{used} = m_{pp} + \int_0^{t_{end}} \dot{m}\,\mathrm{d}t \quad (\times \text{ margin}). \tag{15}$$

6.2 End-of-run constraint

With line losses neglected, the only pressure drop that matters is the minimum the regulator needs to keep regulating at the end-of-run flow:

$$P_b(t_{end}) \geq P_{b,min} \equiv P_t + \Delta P_{reg,min}(\dot{m}). \tag{16}$$

6.3 Minimum volume for a given start pressure

For an adiabatic bottle the end state lies on the start isentrope, $\rho_{b,end} = \rho(P_{b,min}, s_0)$, so

$$\boxed{V_{b,min}(P_{b0}) = \frac{m_{used}}{\rho(P_{b0}, T_{b0}) - \rho(P_{b,min},\, s_0)}} \tag{17}$$

This is implicit: m_{used} depends on $T_{in}(t)$, which depends on how fast P_b and T_b fall, which depends on V_b. It is solved by fixed-point iteration or bisection on V_b, integrating Sections 2–5 each time, for each candidate P_{b0}. The result is a trade curve $V_{b,min}$ versus P_{b0}.

That curve shows diminishing returns at high pressure for two reasons: real-gas $Z > 1$ means density no longer scales with pressure, and a deeper blowdown gives colder gas and more JT cooling.

6.4 Pressure rating

$$MEOP \geq P\big(\rho_{b0},\, T_{max}\big), \qquad P_{burst} \geq SF \cdot MEOP, \tag{18}$$

where ρ_{b0} is the fill density and T_{max} is the highest temperature the bottle can see after fill (hot day, fill heating, soak). The safety factor comes from the governing COPV standard, for example ANSI/AIAA S-081 [4], or the range's rules.

6.5 Temperature constraints

The same model gives the minimum bottle gas temperature $T_b(t_{end})$, which must stay above the liner and composite minimum service temperature. Likewise T_{in} must stay above the regulator seal rating. Adiabatic blowdown from high pressure can violate either before the pressure constraint is reached, so this can drive the size too.

6.6 Mass optimization (optional)

Composite bottle mass scales roughly as

$$W \propto \frac{P_{b0} V_b}{(PV/W)}, \tag{19}$$

where PV/W is the vessel performance factor. Plotting W along the trade curve, with the T_b and regulator limits marked, gives the minimum-mass point that meets every constraint.

6.7 Dispersions and worst case

For the sizing case, take the high end of each input: propellant flow, run time (or full depletion) and collapse factors. For the rating case, take the highest T_{max}. For the volume case, take the lowest start temperature, since a cold fill holds less pressure per unit mass at a given rating.

Nomenclature

Symbol	Meaning
A_i	bottle internal surface area
c_p, c_v	specific heats at constant pressure, volume
$CF_i,\ CF_{pp,i}$	instantaneous collapse factor (expulsion, pre-press), tank i
h	specific enthalpy
h_i	internal heat-transfer coefficient
$\dot{m}$	total pressurant flow leaving the bottle
$\dot{m}_{g,i}$	pressurant flow into tank i
$\dot{m}_{p,i}$	propellant outflow from tank i
$m_b,\ m_{used}$	bottle gas mass; mass removed from bottle
$m_{res,i}$	unusable propellant residual, tank i
$P_b,\ P_t,\ P_u$	bottle, regulated tank, and ullage pressure
$\Delta P_{reg,min}$	minimum regulator pressure drop to stay in regulation
$\dot{Q}_w$	heat transfer from bottle wall to gas
s	specific entropy
$T_b,\ T_{in},\ T_w$	bottle gas, tank inlet, bottle wall temperature
$t_{dep,i}$	depletion time of tank i
$u,\ v$	specific internal energy, specific volume
$V_b,\ V_{u,i},\ \dot{V}_i$	bottle volume; ullage volume; displacement rate
μ_{JT}	Joule–Thomson coefficient
ρ	density

References

[1] L. de Quay and B. K. Hodge, "A History of Collapse Factor Modeling and Empirical Data for Cryogenic Propellant Tanks," NASA Stennis Space Center / Mississippi State University, 2010. NASA NTRS document 20100026018.

[2] R. Span, E. W. Lemmon, R. T. Jacobsen, W. Wagner and A. Yokozeki, "A Reference Equation of State for the Thermodynamic Properties of Nitrogen for Temperatures from 63.151 to 1000 K and Pressures to 2200 MPa," *J. Phys. Chem. Ref. Data*, 29(6):1361–1433, 2000.

[3] I. H. Bell, J. Wronski, S. Quoilin and V. Lemort, "Pure and Pseudo-pure Fluid Thermophysical Property Evaluation and the Open-Source Thermophysical Property Library CoolProp," *Ind. Eng. Chem. Res.*, 53(6):2498–2508, 2014.

[4] ANSI/AIAA S-081, *Space Systems — Composite Overwrapped Pressure Vessels*, American Institute of Aeronautics and Astronautics (use the current revision).